

AI Loan Eligibility Prediction System

Machine Learning Risk Assessment Report

Applicant Name	Anees Ur Rehman
Application ID	APP-FF86F52C
Generated Date & Time	2026-07-03 14:38:43
Best Machine Learning Model Used	XGBoost
Applicant Loan Assessment	Comprehensive AI credit risk evaluation

Generated using Python, Scikit-Learn, Streamlit

Executive Summary

Applicant Risk Summary	Low Risk
AI Decision Summary	APPROVED
Financial Stability Score (0-100)	87
Approval Probability Gauge	■■■■■■■■■■ 87%
One-line AI Verdict	Applicant shows strong financial stability with low predicted credit risk.

Risk Score: ■■■■■■■■■■ 13/100

Risk Interpretation: 0-30 Low Risk | 31-60 Medium Risk | 61-100 High Risk

Applicant shows strong financial stability with low predicted credit risk.

Applicant Profile

Field	Value
■■Applicant Information	
Full Name	Anees Ur Rehman
Application ID	APP-FF86F52C
CNIC / National ID	3810000000000
Phone Number	034589920000
Email Address	Anees@gmail.com
■■Personal Details	
Age	30
Gender	Male
Marital Status	Single
Education Level	Bachelor's
Employment Status	Employed
■■Financial Details	
Annual Income	\$50,000.00
Monthly Income	\$5,000.00
Credit Score	650
Debt To Income Ratio	30.0%
■■Loan Details	
Loan Amount	\$10,000.00
Loan Purpose	Business
Grade / Subgrade	A1
Interest Rate	10.0%
Loan Term	36 Months

AI Decision Report

LOAN STATUS: APPROVED

Loan Status	APPROVED
Approval Probability	87.32%
Financial Risk Score (0-100)	13
Risk Level	Low
Model Used	XGBoost
Prediction Date	2026-07-03 14:38:43

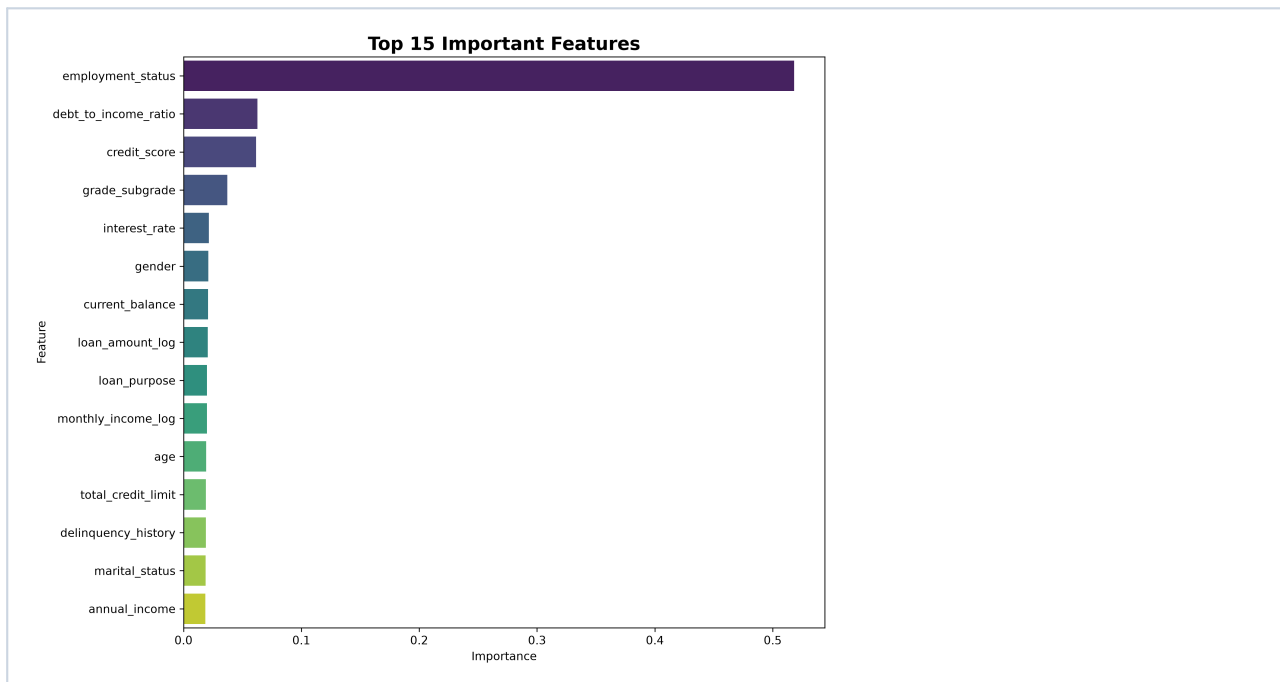
Risk Score: 13/100

0-30 Low Risk | 31-60 Medium Risk | 61-100 High Risk

AI Insight Section

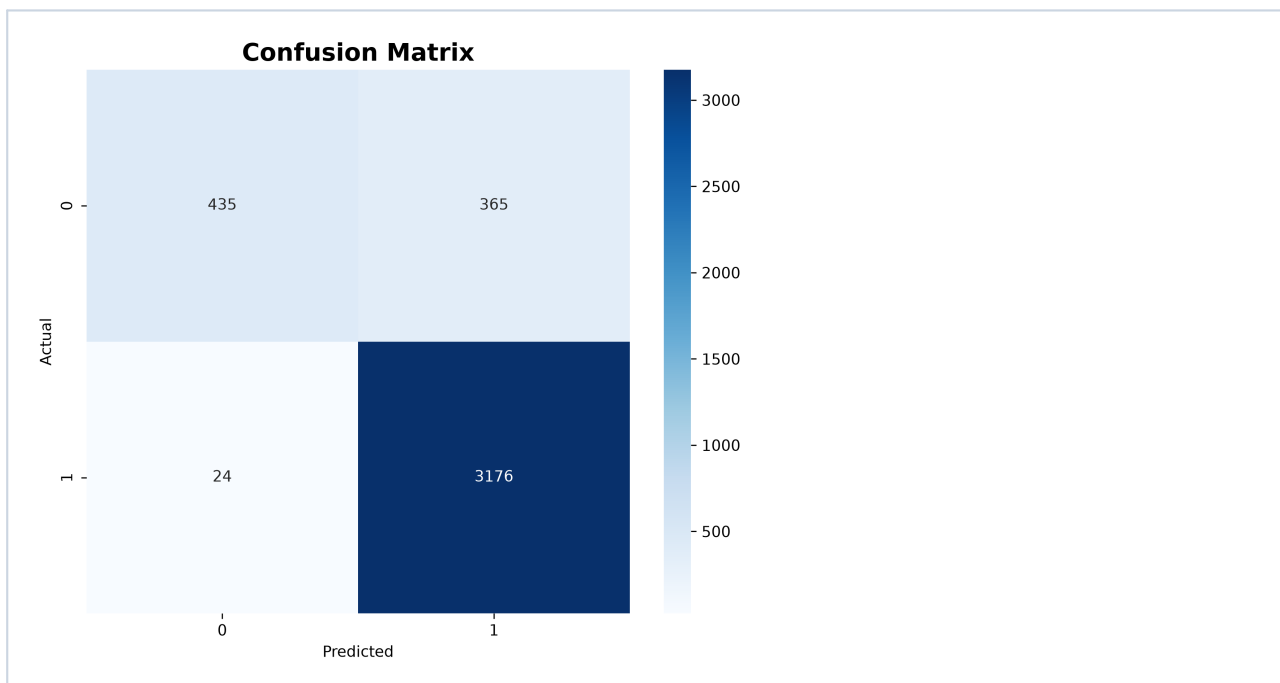
- Income-to-obligation balance indicates strong repayment readiness.
- Credit profile reflects disciplined payment behavior and healthy utilization.
- Debt exposure remains within acceptable underwriting thresholds.
- Employment and financial stability support long-term repayment consistency.
- Overall borrower profile aligns with prudent bank lending standards.

Feature Importance



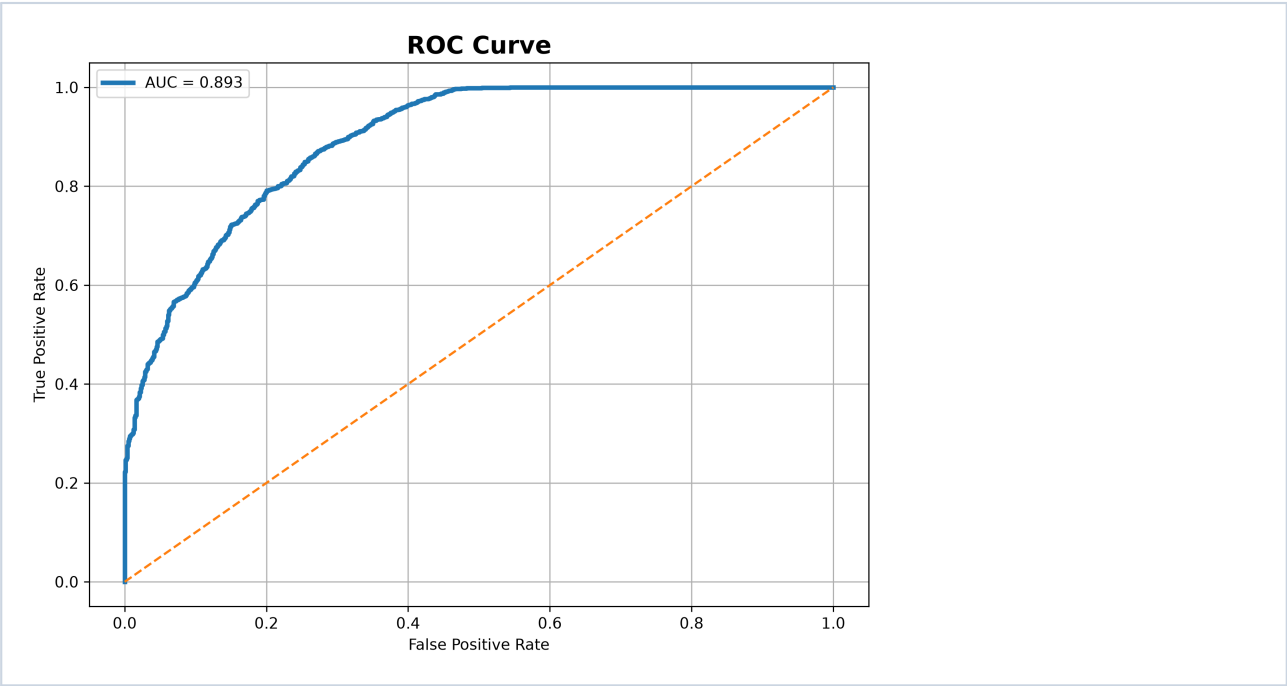
Relative influence of key applicant attributes on model decision.

Confusion Matrix



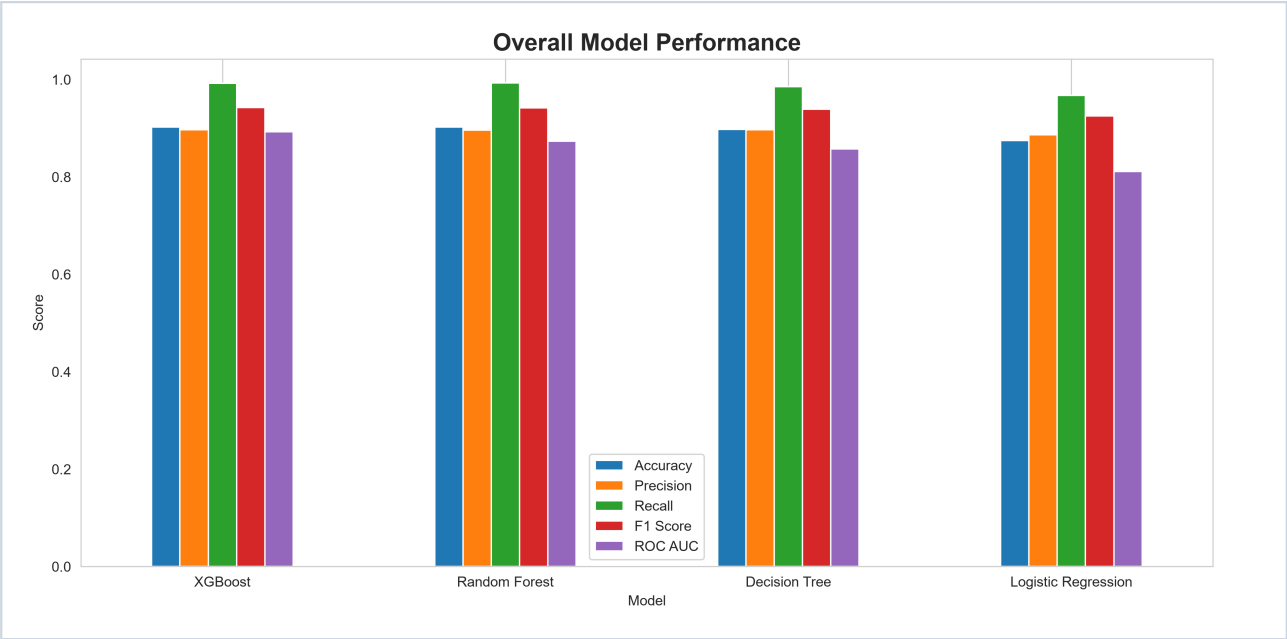
Distribution of true vs predicted classifications.

ROC Curve



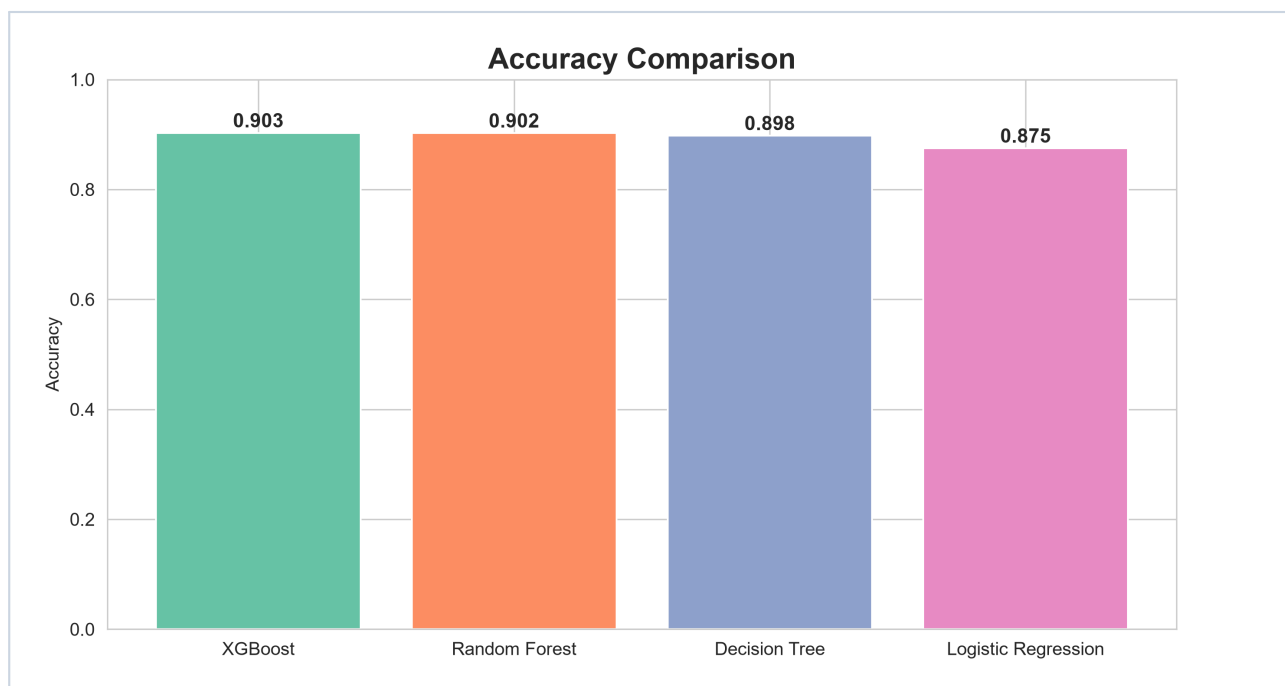
Model discrimination performance across threshold settings.

Model Dashboard



Consolidated overview of model diagnostics and performance.

Accuracy Comparison



Comparative accuracy across evaluated machine learning models.

Enhanced Model Comparison

Rank	Model	Accuracy	Precision	Recall	F1 Score
1	XGBoost	0.9028	0.8969	0.9925	0.9423
2	Random Forest	0.9025	0.8964	0.9928	0.9422
3	Decision Tree	0.8978	0.8968	0.9856	0.9391
4	Logistic Regression	0.875	0.8866	0.9675	0.9253

Executive Banking Final Summary

Best Model Name	XGBoost
Accuracy	0.9028
Risk Level	Low
Financial Score	13 / 100
Final Decision	APPROVED

Scan to view project source code

QR module unavailable. QR section skipped safely.

Approved by AI Credit Intelligence System

Signature: _____

Timestamp: 2026-07-03 14:38:43

System Version: v2.0